



Department of Textile and Fibre Technology
Indian Institute of Technology, Delhi

Major Project Part I (TXD 401)

Mid-Infrared Imaging and Machine Learning Approaches for Non-Destructive and Efficient Material Detection of Textile Fabrics

Lakshya Kumar | 2022TT12183

Keshav Chachan | 2022TT12148

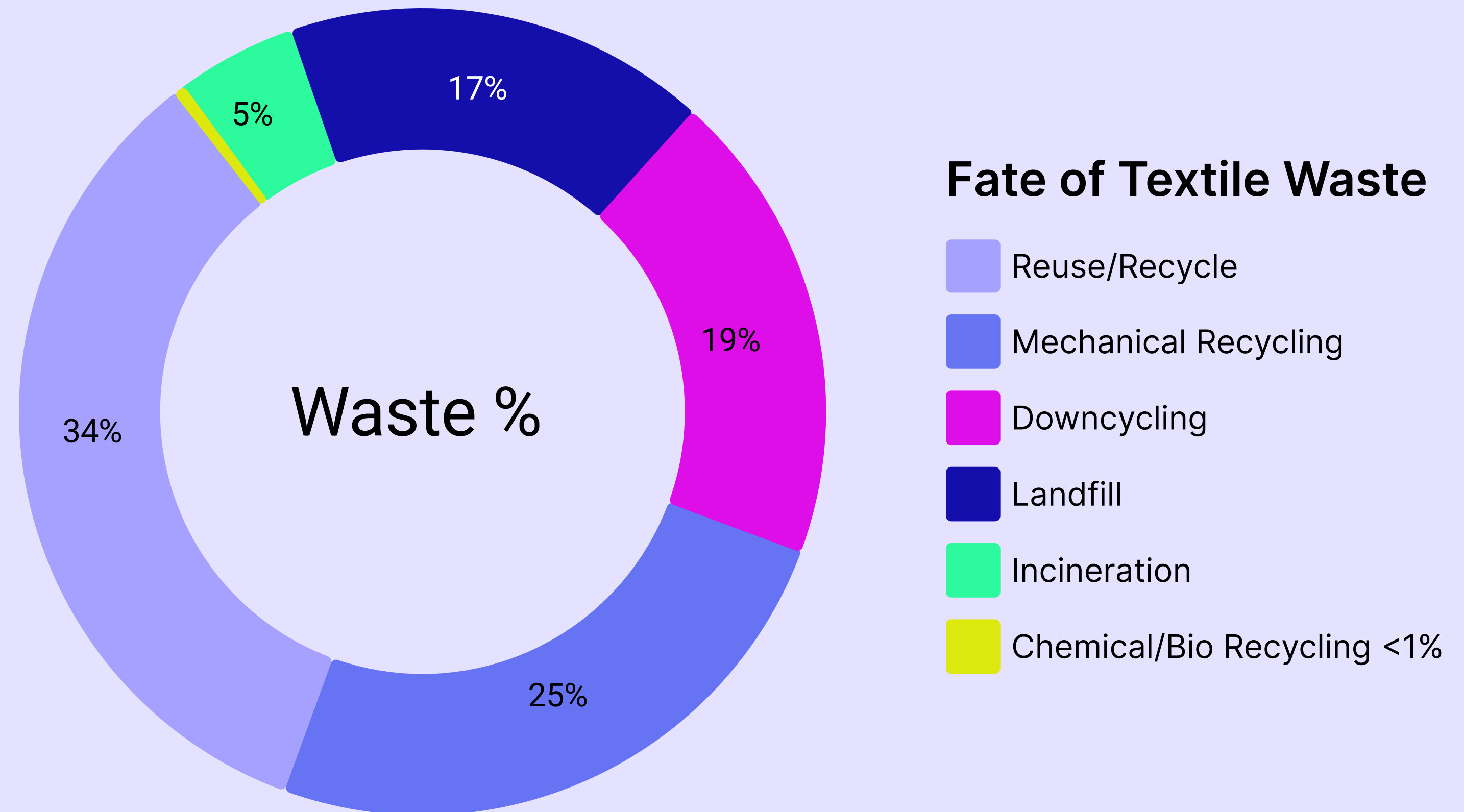
Supervisor : Prof. Apurba Das





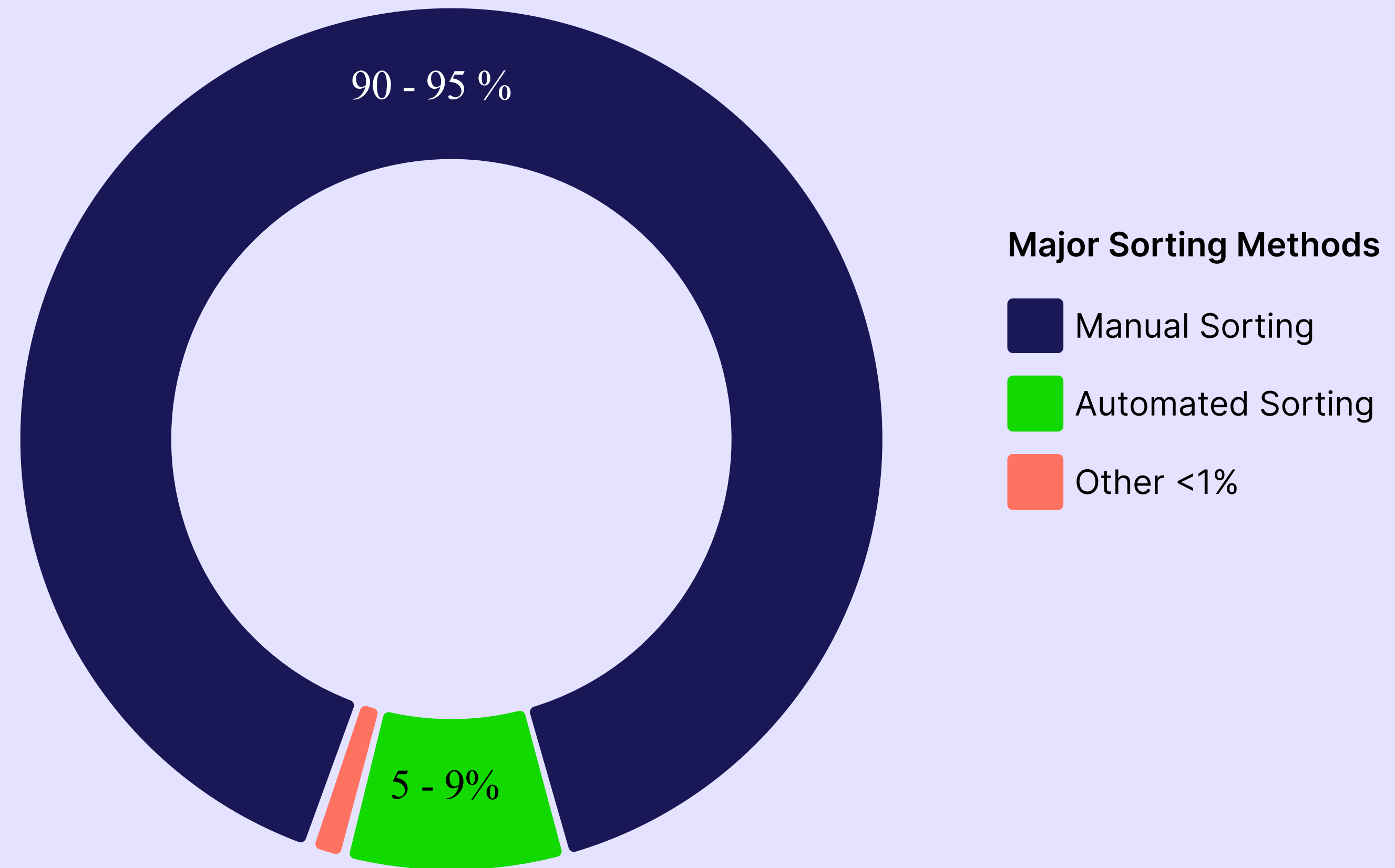
“Do you know that in India, 7.8 million tons of textile waste is generated annually”

What happens with those textile waste in India?



Source : <https://reports.fashionforgood.com/report/sorting-for-circularity-india-wealth-in-waste/chapterdetail?reportid=813&chapter=3>

Distribution of Major Sorting Methods in India



Source : <https://reports.fashionforgood.com/report/sorting-for-circularity-india-wealth-in-waste/chapterdetail?reportid=813&chapter=3>

Literature Review

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Figure 1. A photograph of Marena Manley, a woman with dark hair and glasses, wearing a red necklace.

Marena Manley was born in 1966 and grew up in the Northern Cape, South Africa. She obtained her BSc in Food Science degree at Stellenbosch University and subsequently her Honours and MSc degrees at the University of Pretoria. She received her PhD (in near-infrared spectroscopy) in 1995 from the University of Plymouth, UK. After 18 months in industry, she joined Stellenbosch University in 1997 as a lecturer where she is currently appointed as Professor. Her research interests involve the application of near-infrared (NIR) spectroscopy, NIR hyperspectral imaging and lately also X-ray micro-computed tomography to study whole grain endosperm texture and other cereal defects.

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Chem. Soc. Rev., 2014, 43, 8200–8214 | 8201

Fundamental principles of near-infrared spectroscopy

Spectra in the NIR region result from energy absorption by organic molecules, and comprise overtones and combinations of overtones originating from fundamental bond vibrations (stretching or bending) occurring in the mid-infrared (MIR) region of the spectrum. The features in NIR spectra of organic compounds are thus orders of magnitude weaker than those in the MIR. Overtones can be found by dividing the wavelengths in the infrared region by approximately 2, 3 or 4 and provide the advantage of a dilution series. NIR spectra can thus be collected directly on samples without the need for dilution, enabling direct analysis of solid samples. Ease of sample preparation and presentation results in NIR spectroscopy often being used in favour of MIR spectroscopy.

The NIR region extends from 800 to 2500 nm (12 500 to 4000 cm^{-1} ; 120 to 375 THz), between the visible from 380 to 780 nm (26 316 to 12 820 cm^{-1} ; 385–790 THz) and MIR from 2500 to 15 000 nm (4000 to 400 cm^{-1} ; 30 to 120 THz) regions. NIR spectra contain information about the major X–H chemical bonds, i.e. C–H, O–H and N–H. All molecules containing hydrogen will have a measurable NIR spectrum, resulting in a large range of organic materials to be suitable for NIR analysis.

Due to the overtone and combination modes and large numbers of possible vibrations, NIR spectra are very complex, consisting of many overlapping peaks (referred to as 'multicollinearity'), which result in broad bands. This makes it difficult to interpret NIR spectra visually, assign specific features to specific chemical components or extract information contained in the spectra easily. It was, however, realised early on that, with the use of appropriate regression techniques, relationships between absorption values at specific wavelengths and reference values of the constituent to be predicted could be established. Specific chemical

The NIR region extends from 800 to 2500 nm (12 500 to 4000 cm^{-1} ; 120 to 375 THz), between the visible from 380 to 780 nm (26 316 to 12 820 cm^{-1} ; 385–790 THz) and MIR from 2500 to 15 000 nm (4000 to 400 cm^{-1} ; 30 to 120 THz) regions.

Due to the overtone and combination modes and large numbers of possible vibrations, NIR spectra are very complex, consisting of many overlapping peaks (referred to as 'multicollinearity'), which result in broad bands. This makes it difficult to interpret

Vibrational Transitions

- Fundamental Band: Transition from the ground vibrational state ($v=0$) to the first excited state ($v=1$).
- Overtone Band: Transition from the ground vibrational state ($v=0$) to a higher excited state ($v=2$, $v=3$, etc.).
- Combination Band: Simultaneous excitation of two or more different fundamental vibrations.

Literature Review

THz Frequency Range

0.1–10 THz

Equivalent to 3.3–333 cm^{-1} (far-infrared region)

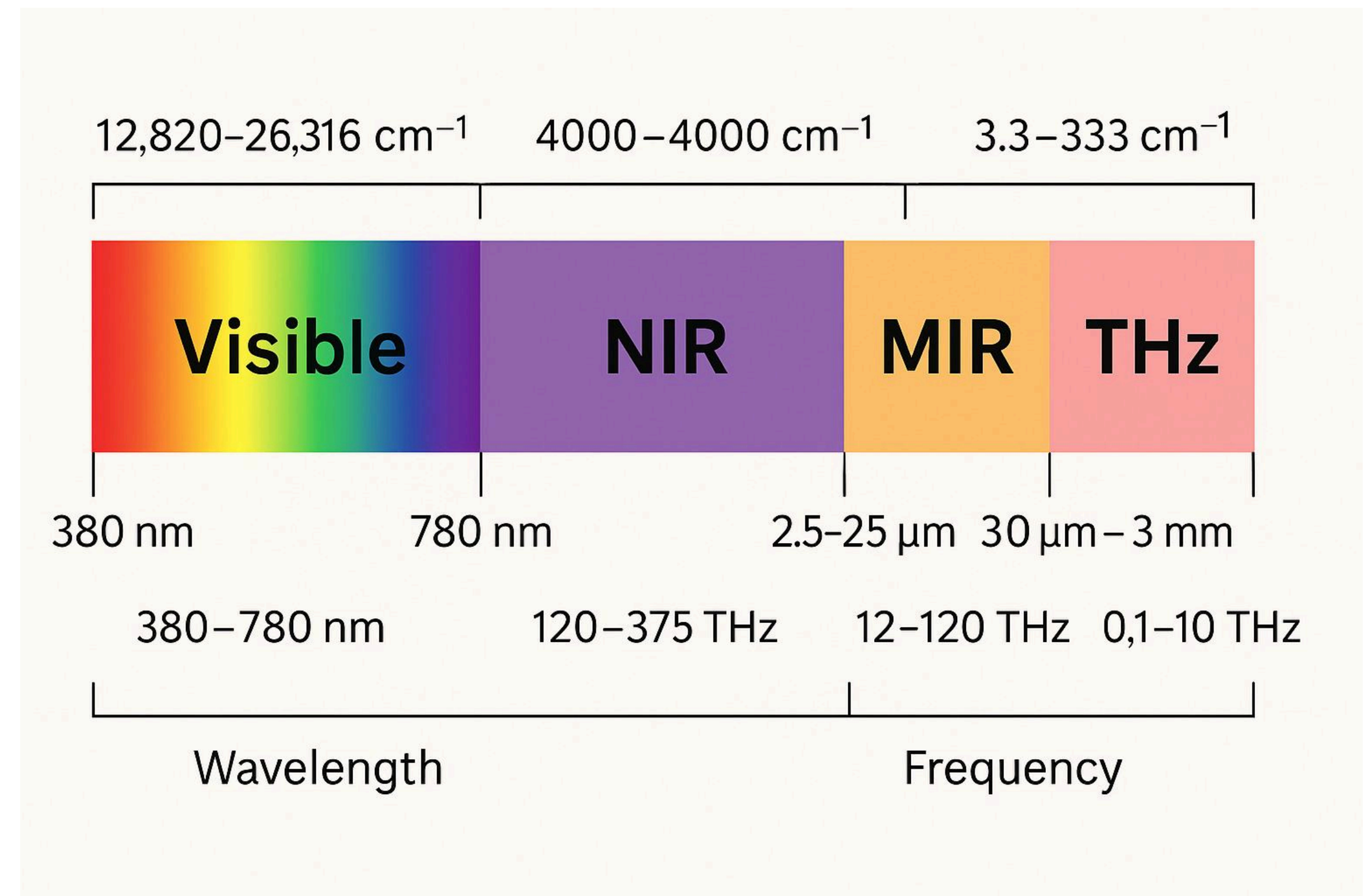
Wavelength range: 30 μm – 3 mm.

What THz Detects (Fundamentals)

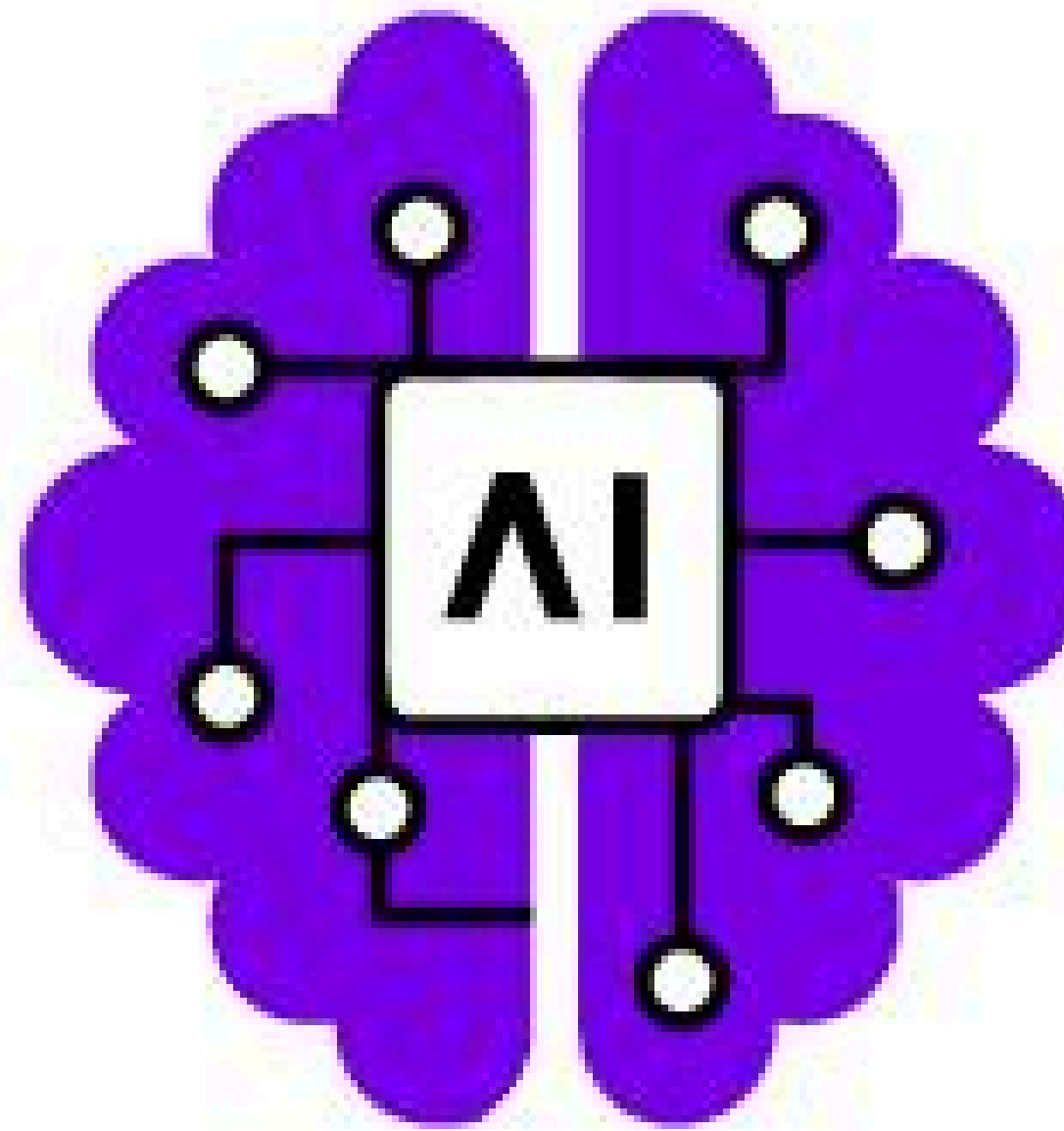
Low-energy molecular vibrations

Intermolecular modes (hydrogen-bond networks, van der Waals forces)

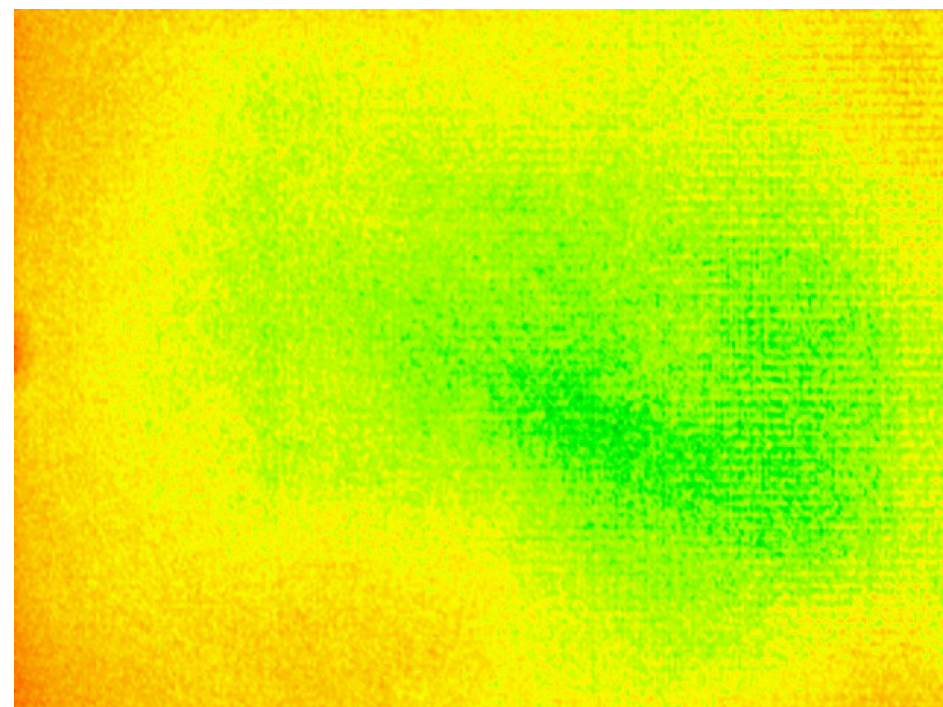
Structural differences in fibers (crystallinity, chain packing)



Input



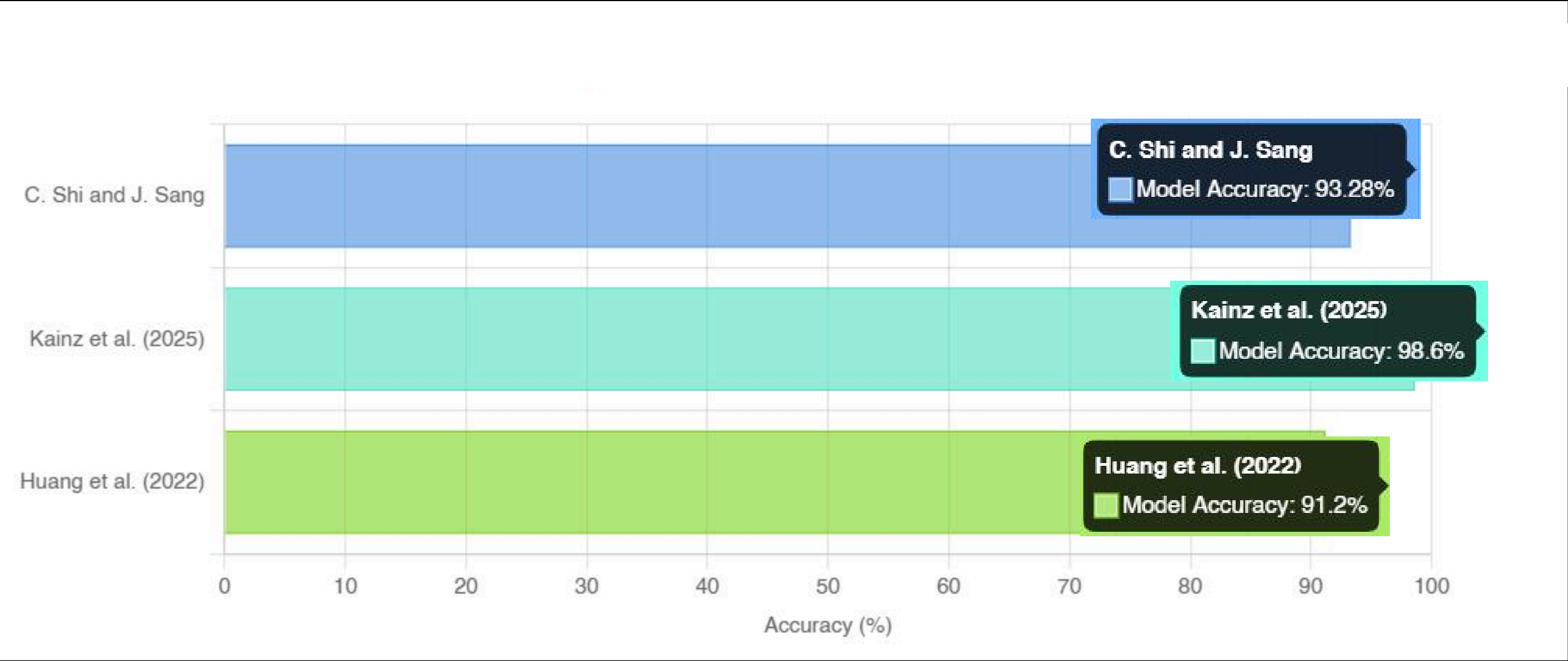
Output



(Cotton)

Literature Review

Reported Accuracy Levels of Automated Textile Sorting in Literature



Paper	Data Source	Total Spectra Sample	Source Categories
Shi & Sang (2025)	Created their own	11,147	18 (pure & blends)
Huang et al. (2022)	Commercial fibers	600,404	25 (natural & synthetic)
Kainz et al. (2025)	Created their own	Not specified	12 (pure & blends)

Literature Review

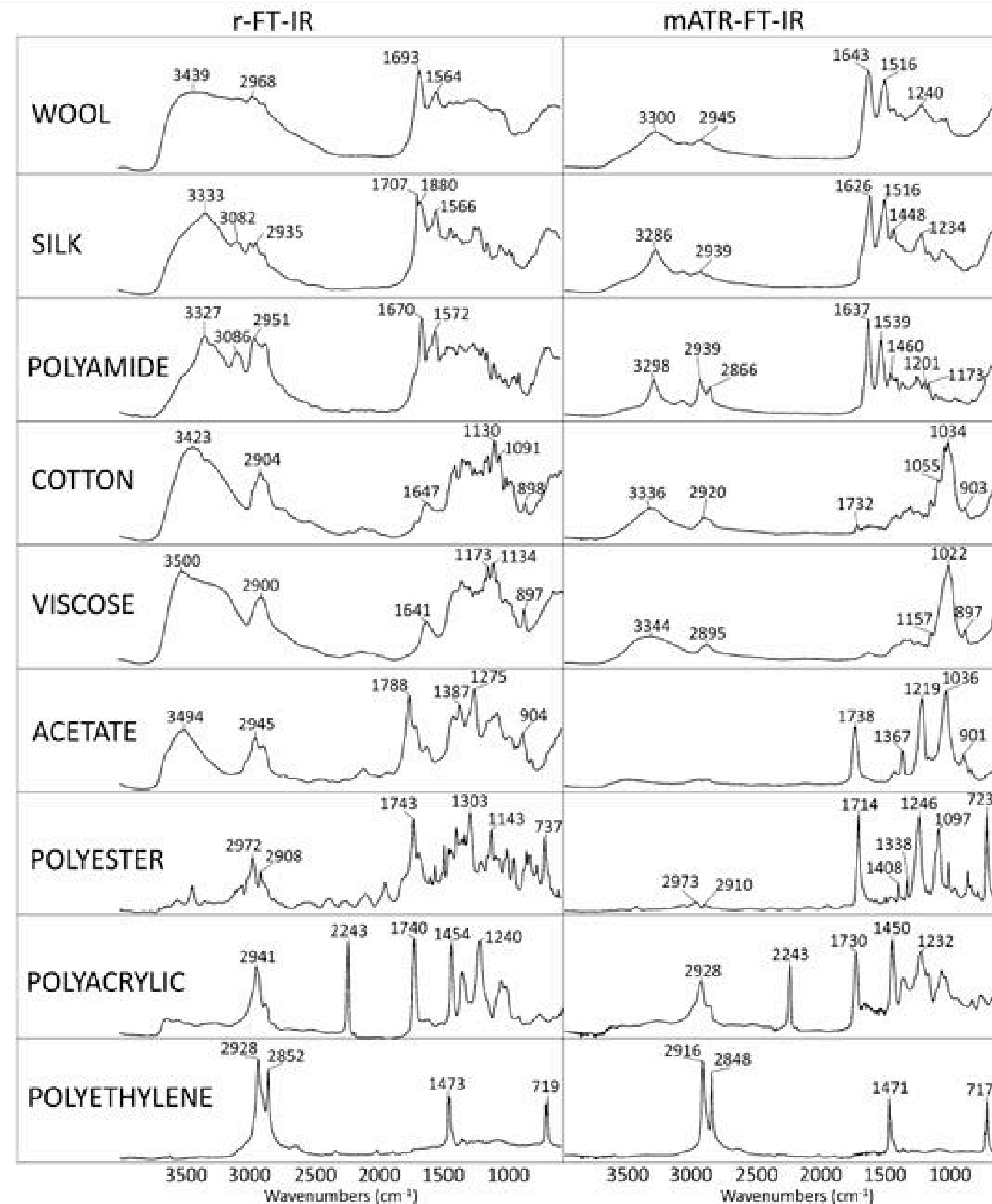


Fig. 1 r-FT-IR and mATR-FT-IR spectra of most common textile fibers. Spectra in figure represent normalized and averaged results of each class. The vertical axis represents absorbance. As natural cellulose-based fibers have very similar spectra, only cotton is presented. Spectra of other natural fibers like linen, jute and sisal are shown in Additional file 1

Fundamental Vibrations by using FT-IR spectroscopy of textile materials

The peaks lie in the range of MIR

Hypothesis

Research Gap

Limited Specificity from Weak NIR Signals:

The inherently weak intensity and broadness of NIR overtone/combination bands significantly limit specific material identification, unlike the strong, specific signals of MIR fundamentals.

Very Limited Research in THz Spectroscopy for Textiles:

THz studies are scarce and mostly limited to pristine, single-fiber lab samples (cotton, wool, polyester, silk), with almost no work on dyed, blended, multilayer, or post-consumer textiles

Research Hypothesis

“Is there any relation between MIR images and the material of textile fabric?”

“Can we use THz spectroscopy to differentiate between fabrics with different dyes”

Objectives

- Verify sample labels using burn tests and solubility tests.

- Train YOLO models on MIR data to evaluate classification accuracy.

- Identify MIR limitations caused by dyes, finishes, and texture effects.

- Identify differentiating spectral peaks from THz data and hyperspectral images.

- Use these extracted peaks to define the THz experimental plan for textile identification

Methodology (Research Framework)

Fabrics Chosen

3
types

Cotton | Polyester | Nylon

Woven and Knitted both were considered

Dataset Size

120
Images

Each image will be in the file formats .png
or .jpeg

Equipment Required

MIR camera (wavelength 7.5 μ m- 14 μ m)



Experimental Section

Experimental Verification of Fabric Material

Objective: To confirm the identity of unknown fabric samples using physical (burn) and chemical (solubility) tests.

Fibre	Reaction to Flame	Odour	Ash/Residue
Cotton	Burns quickly without melting. Continues to glow after flame is removed.	Burning Paper	Leaves a small, fluffy, grey ash.
Nylon	Fuses, shrinks, and melts. Usually self-extinguishing.	(No distinct odour listed, often celery-like)	Leaves a hard, tough, grey or brown bead.
Polyester	Fuses, shrinks, and melts with black smoke. Usually self-extinguishing.	Chemical	Leaves a hard, tough, round black bead.



Supposed to be nylon but blend



Polyester

Experimental Section

Experimental Verification of Fabric Material Chemical Tests

Cotton

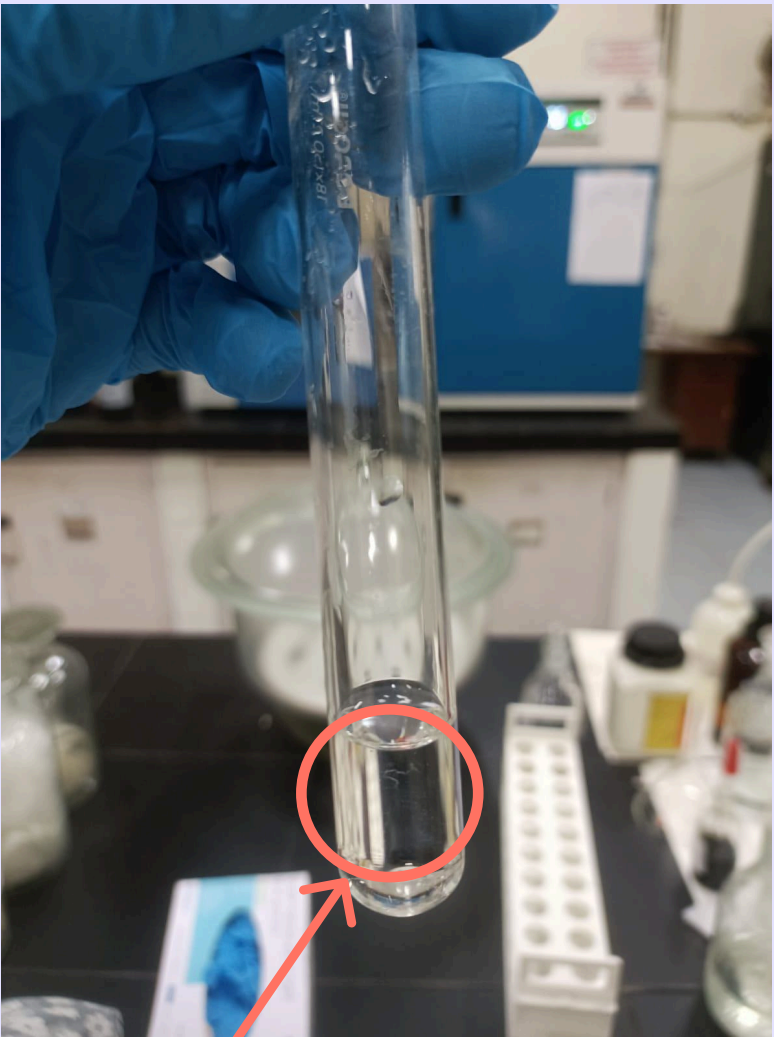
Acetone	Insoluble
5N HCl	Insoluble
59% H ₂ SO ₄	Insoluble
75% H ₂ SO ₄	Soluble

Nylon

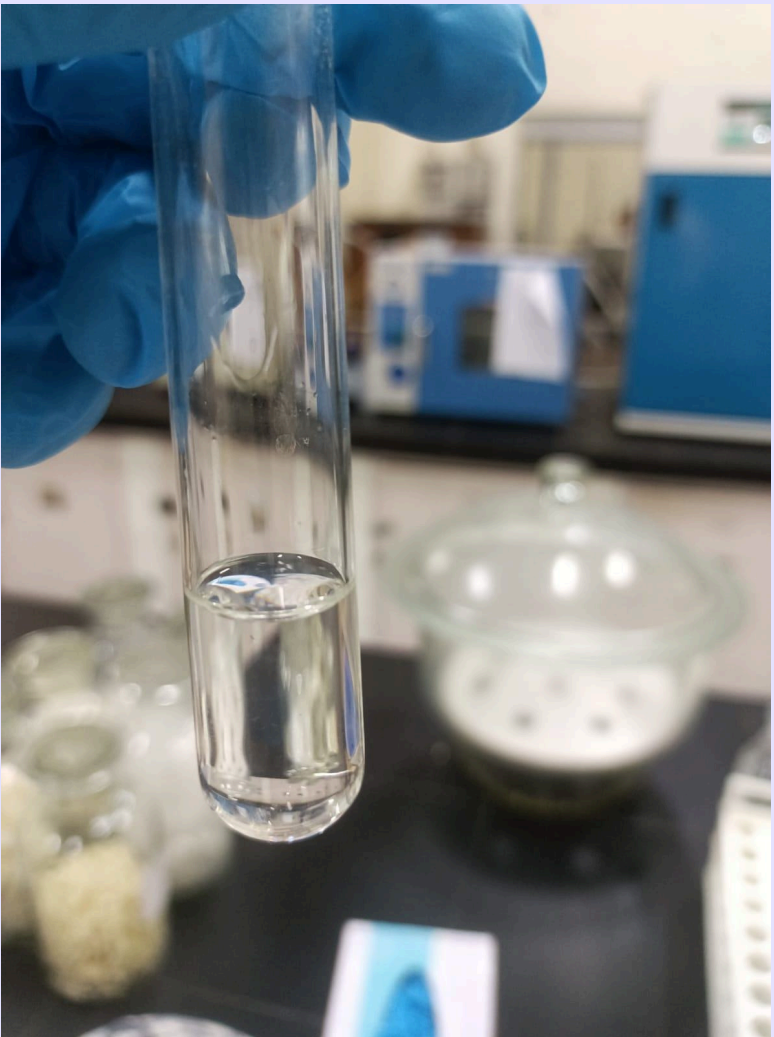
Acetone	Insoluble
5N HCl	Soluble

Polyester

Acetone	Insoluble
5N HCl	Insoluble
59% H ₂ SO ₄	Insoluble
75% H ₂ SO ₄	Insoluble
Conc. H ₂ SO ₄	Soluble



Cotton Sample



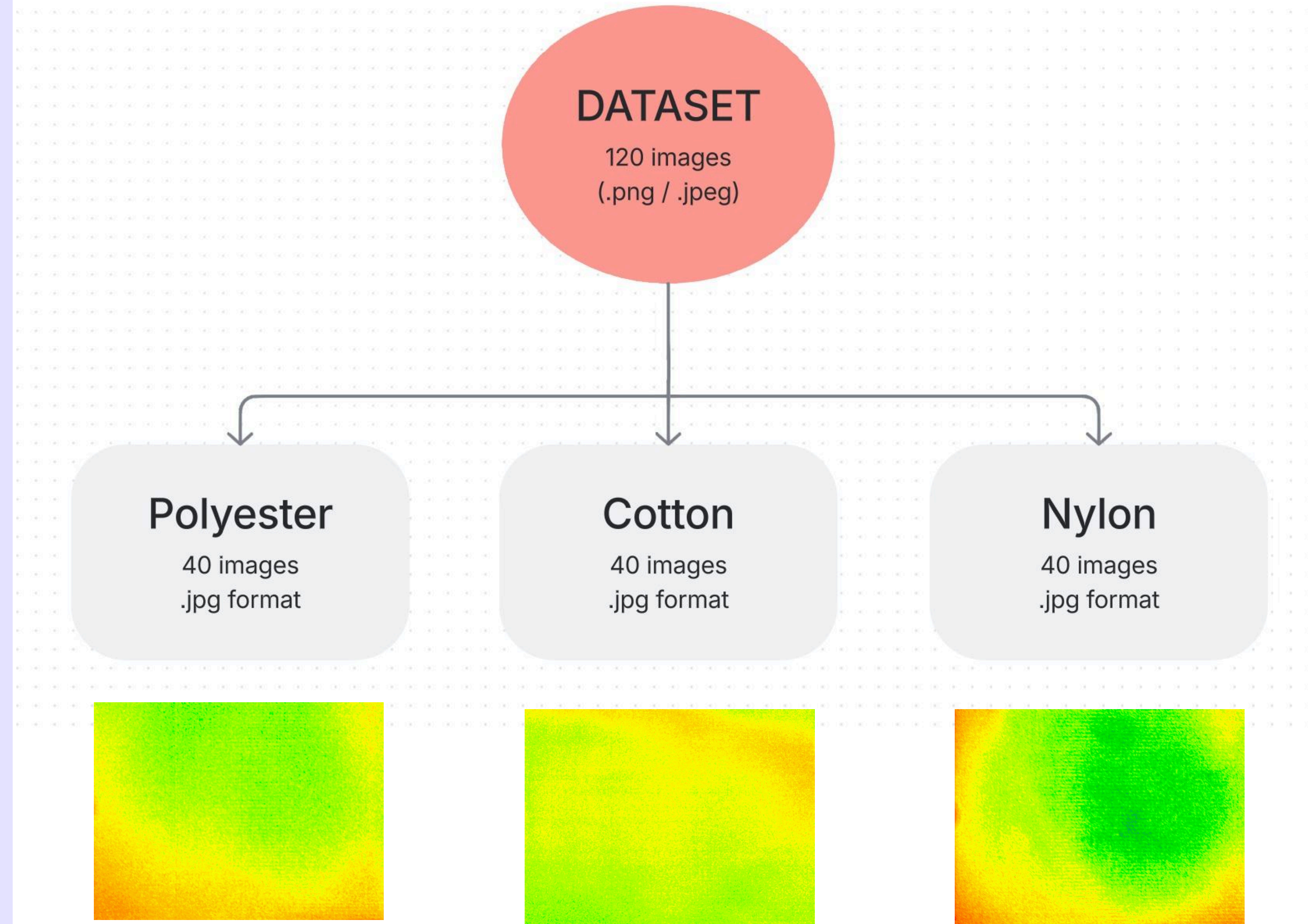
Cotton Sample Dissolved

Dataset Overview

MIR Image

Dimensions : 1280 x 960
Width : 1280 pixels
Height : 960 pixels

- To ensure consistent scale and focus across all samples, a standardized distance of **0.07 meters** was chosen between the camera lens and the sample.
- Throughout the experiment, the environmental temperature was maintained within a controlled range of **21°C to 24°C**.



Phase I

Training Results based on MIR

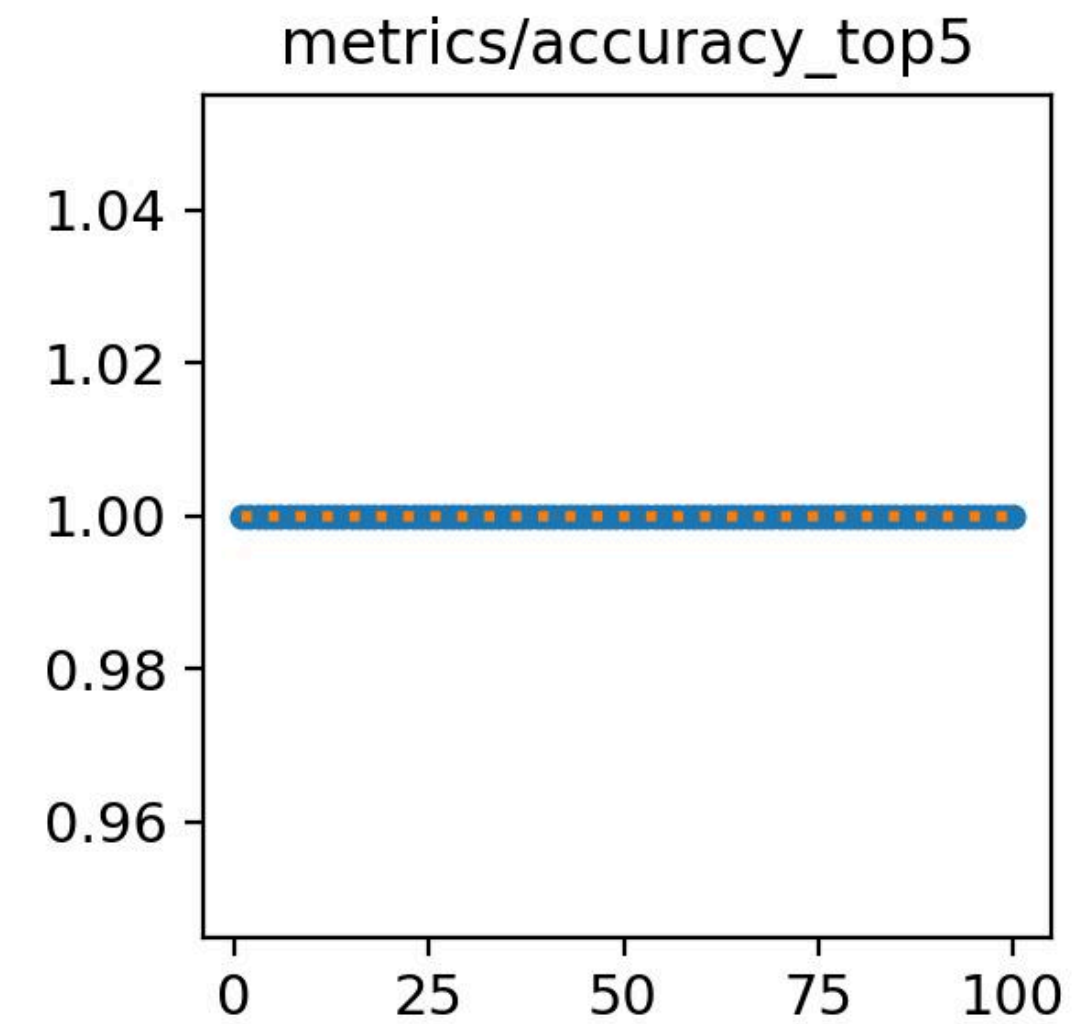
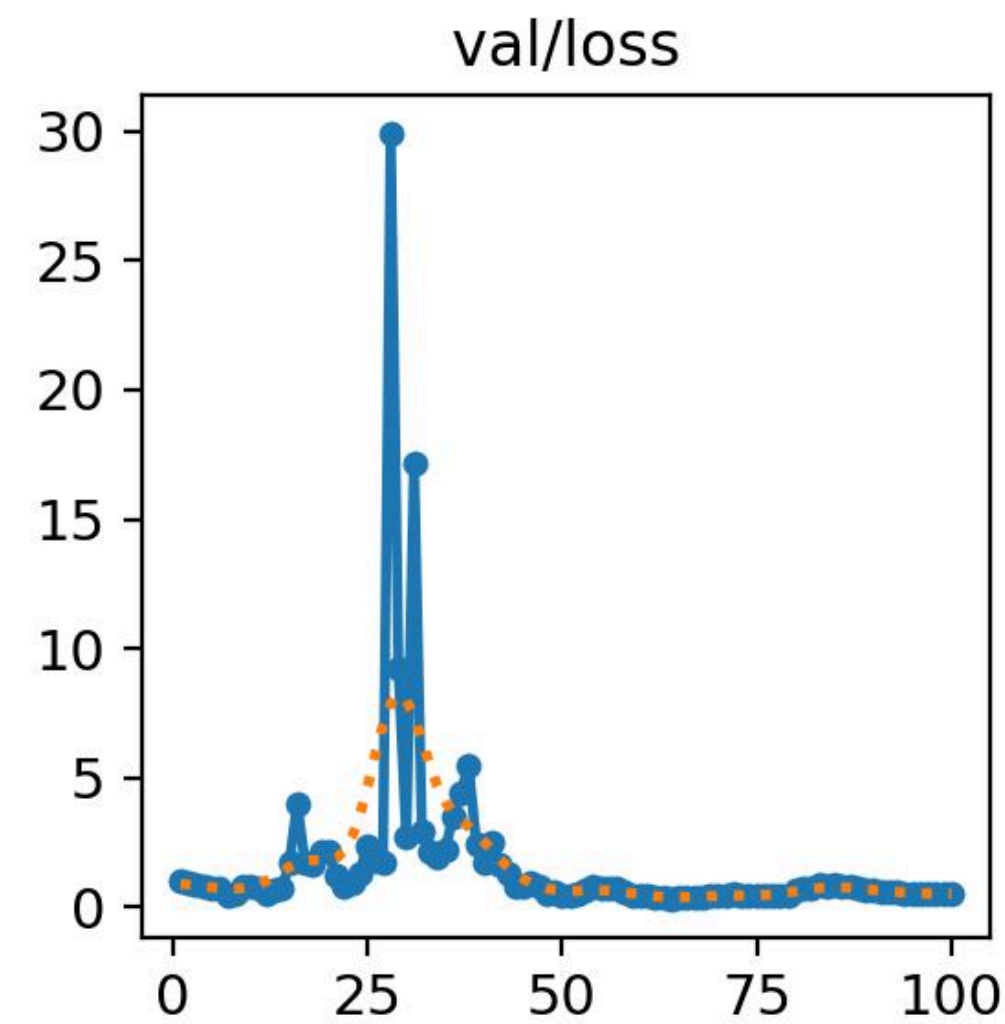
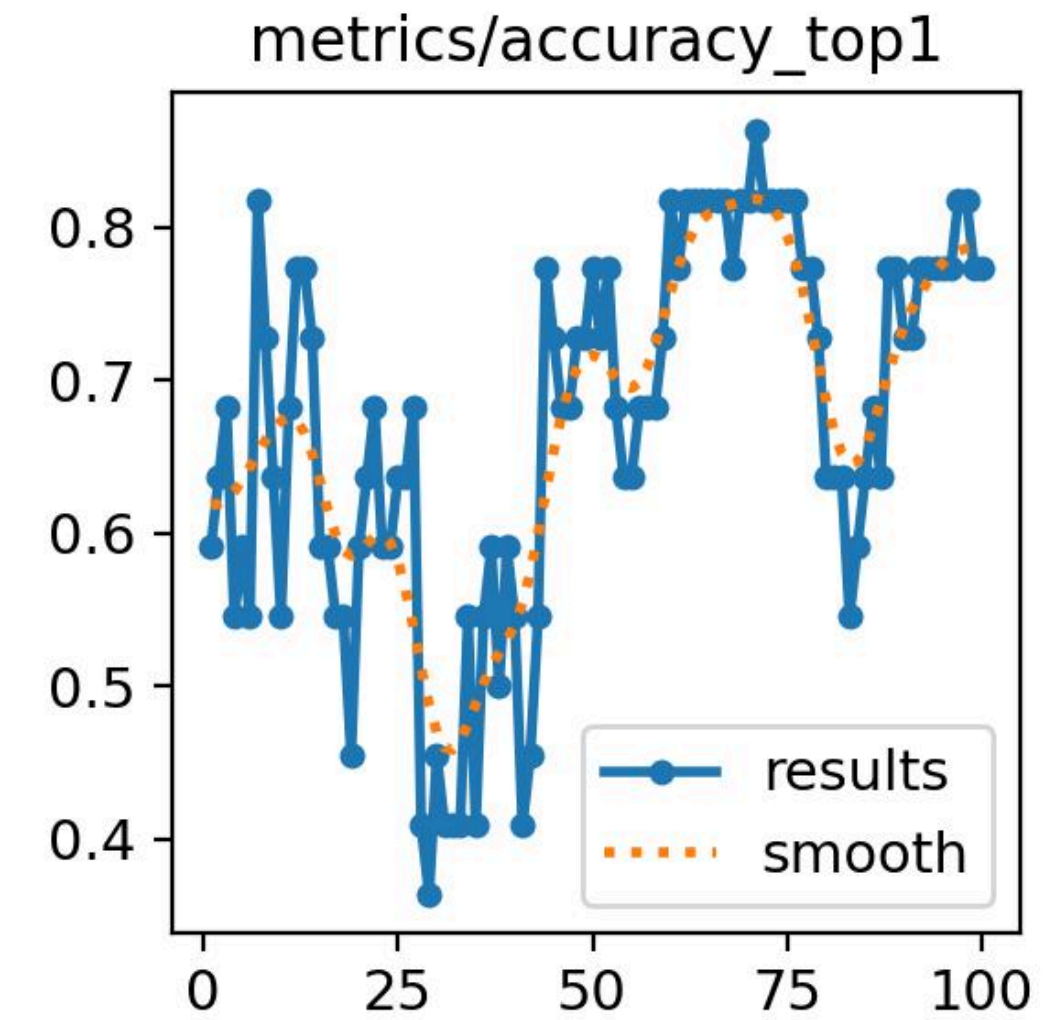
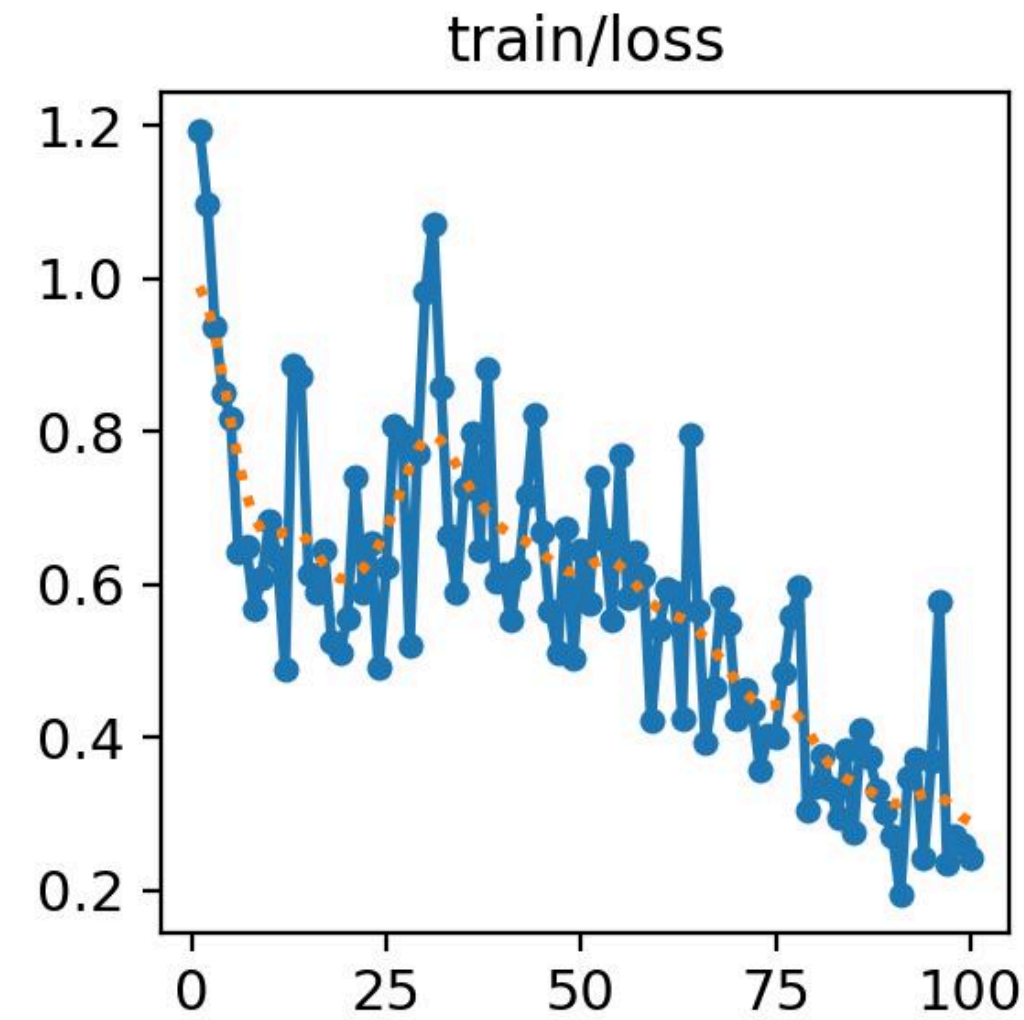
Model Architecture:
YOLO11x-cls initialized with pretrained weights.

Training Parameters:
100 Epochs, Batch size 16.

Augmentation: Horizontal flip, mosaic, and random erasing.

Validation Results:
High accuracy on controlled validation set.

Confusion Matrix: 1.00 accuracy for Polyester/Cotton; slight confusion between Nylon/Cotton.



Phase I

Training Results based on MIR

Model Architecture:
YOLO11x-cls initialized with pretrained weights.

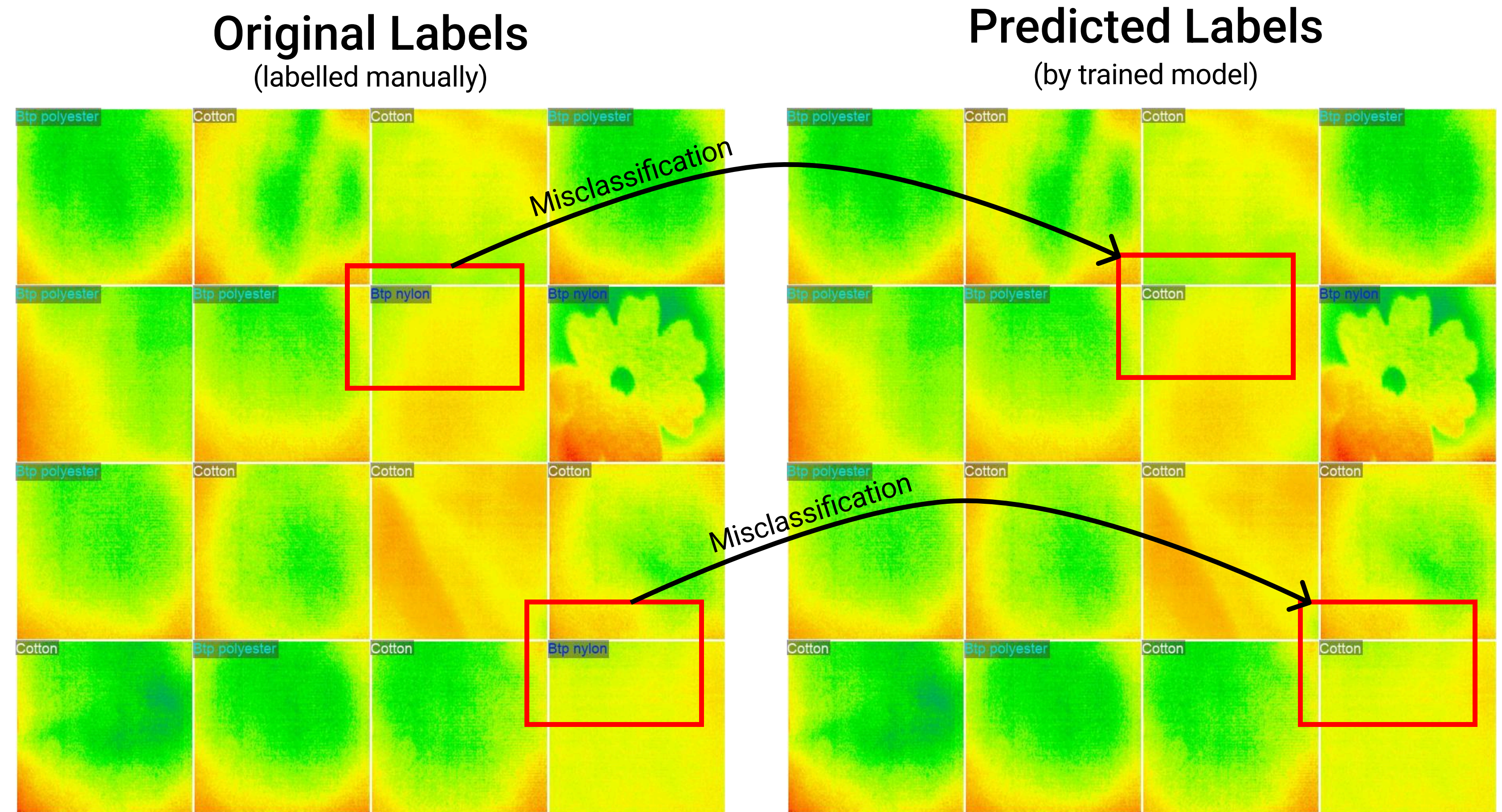
Training Parameters:
100 Epochs, Batch size 16.

Augmentation: Horizontal flip, mosaic, and random erasing.

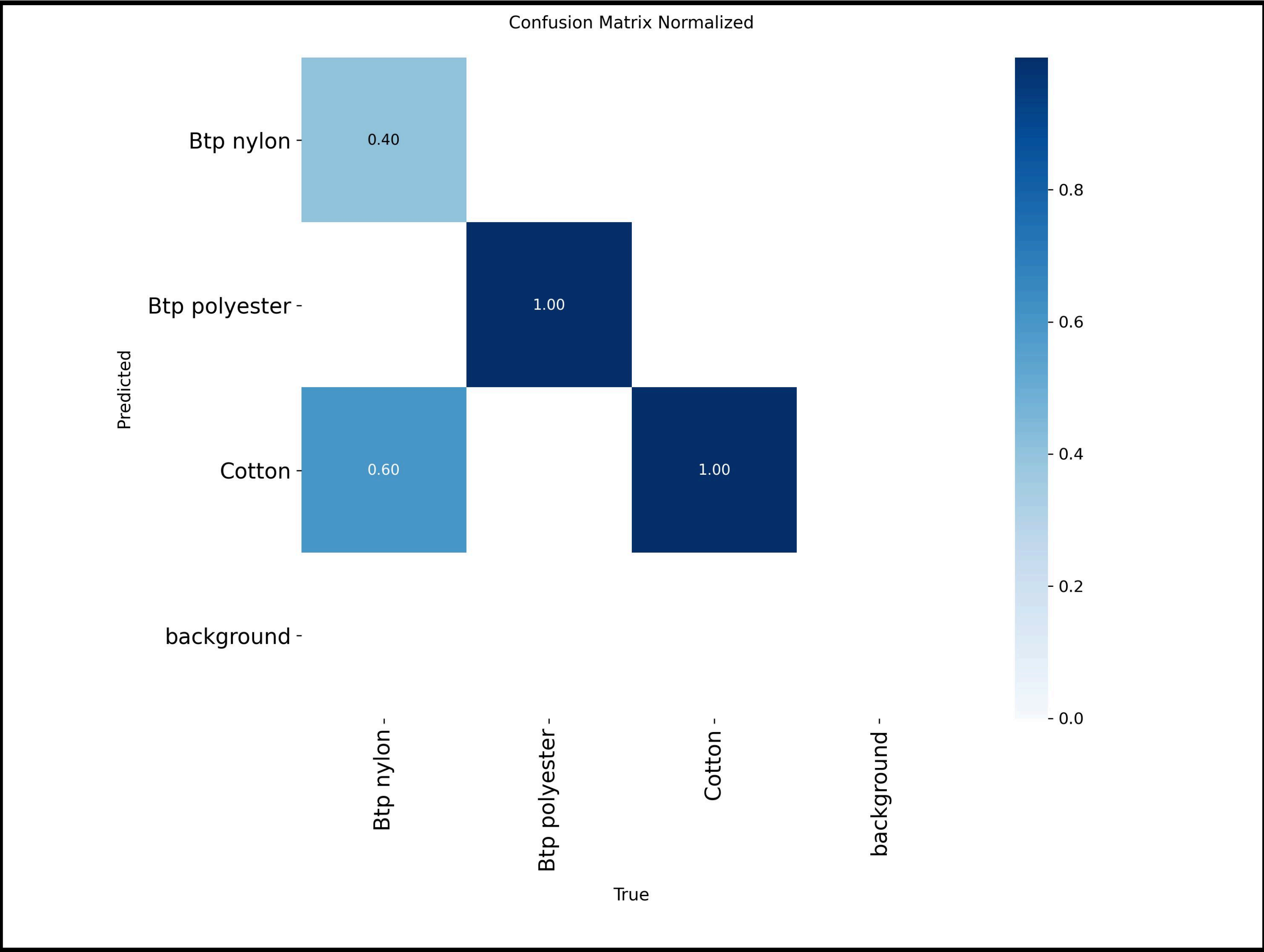
Validation Results:
High accuracy on controlled validation set.

Confusion Matrix: 1.00 accuracy for Polyester/Cotton; slight confusion between Nylon/Cotton.

Visually explained results for trained model



Confusion Matrix



Limitations Identified

MIR Failure Cases: Model lacked robustness in uncontrolled environments.

Key Limitations Identified:

- Thermal Noise: Passive MIR is sensitive to background radiation.
- Surface Dependency: Highly affected by dye concentration and fabric texture.
- The Solution: Transition to Terahertz (THz) Spectroscopy.
- Why? Deeper penetration and resistance to dye interference.

Phase II

THz Dye

Independence Study

A total of **6 cotton-based samples** were prepared for THz analysis to represent various real-world conditions:

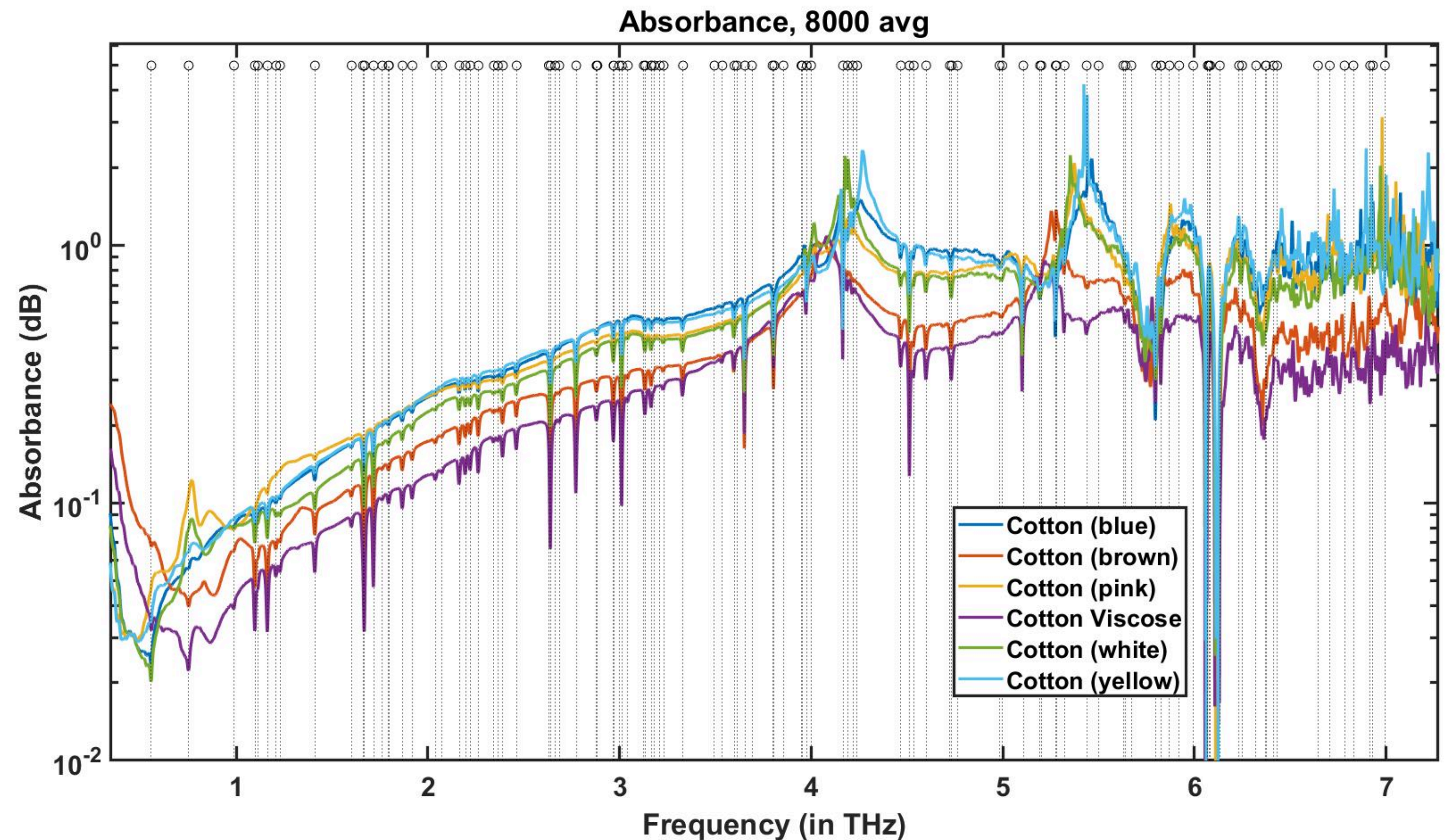
1. Laboratory-Dyed Samples:

Three standard cotton samples were individually dyed in the laboratory using three different classes of dyes: Direct dye, Vat dye, and Reactive dye.

2. Commercial Samples:

Three samples were procured from the market:

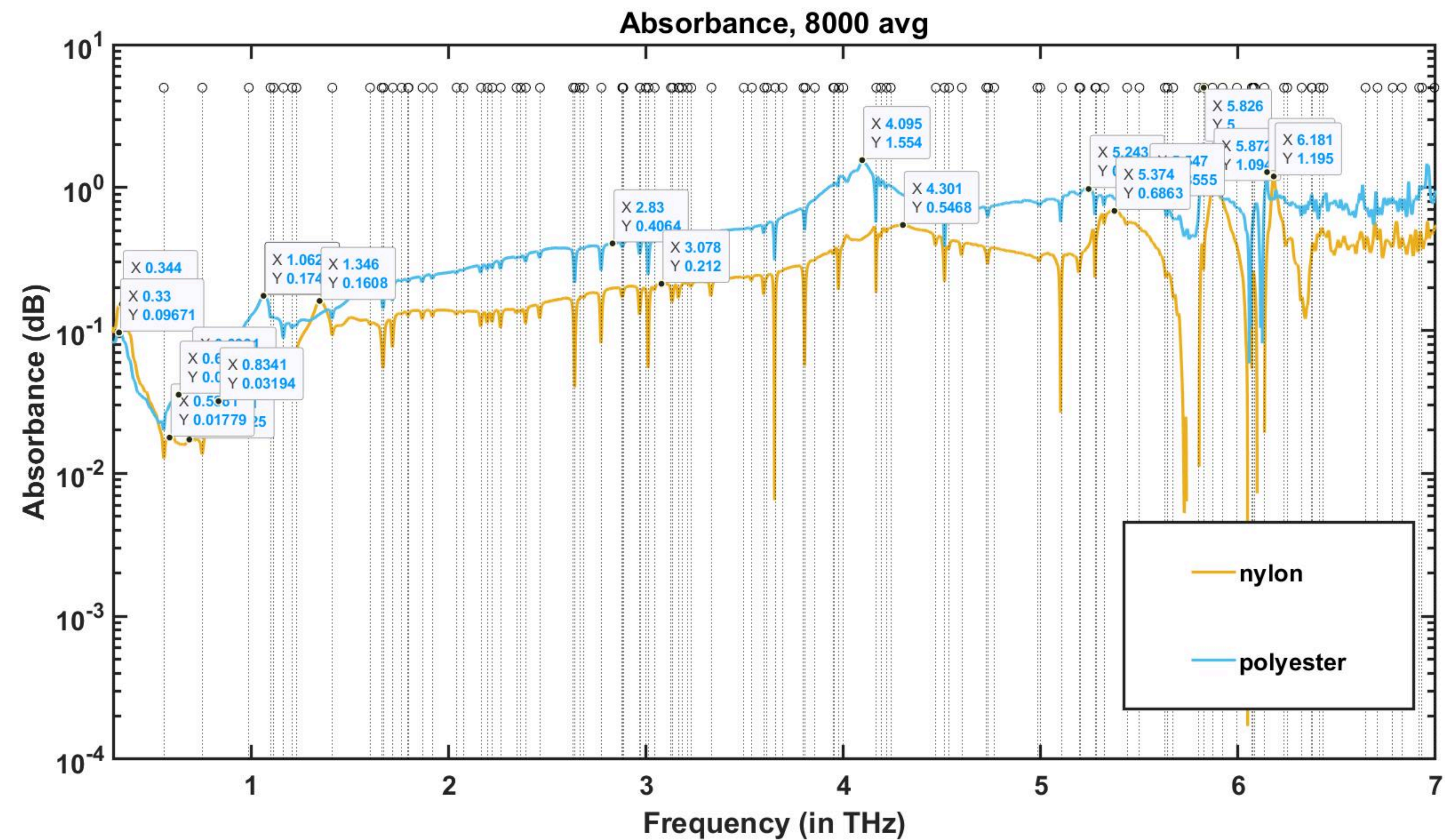
- standard White Cotton
- commercially dyed Pink Cotton
- a Cotton-Viscose blend



Terahertz Time-Domain Spectroscopy (0.5 - 7.5 THz) with 8000 averages per scan.

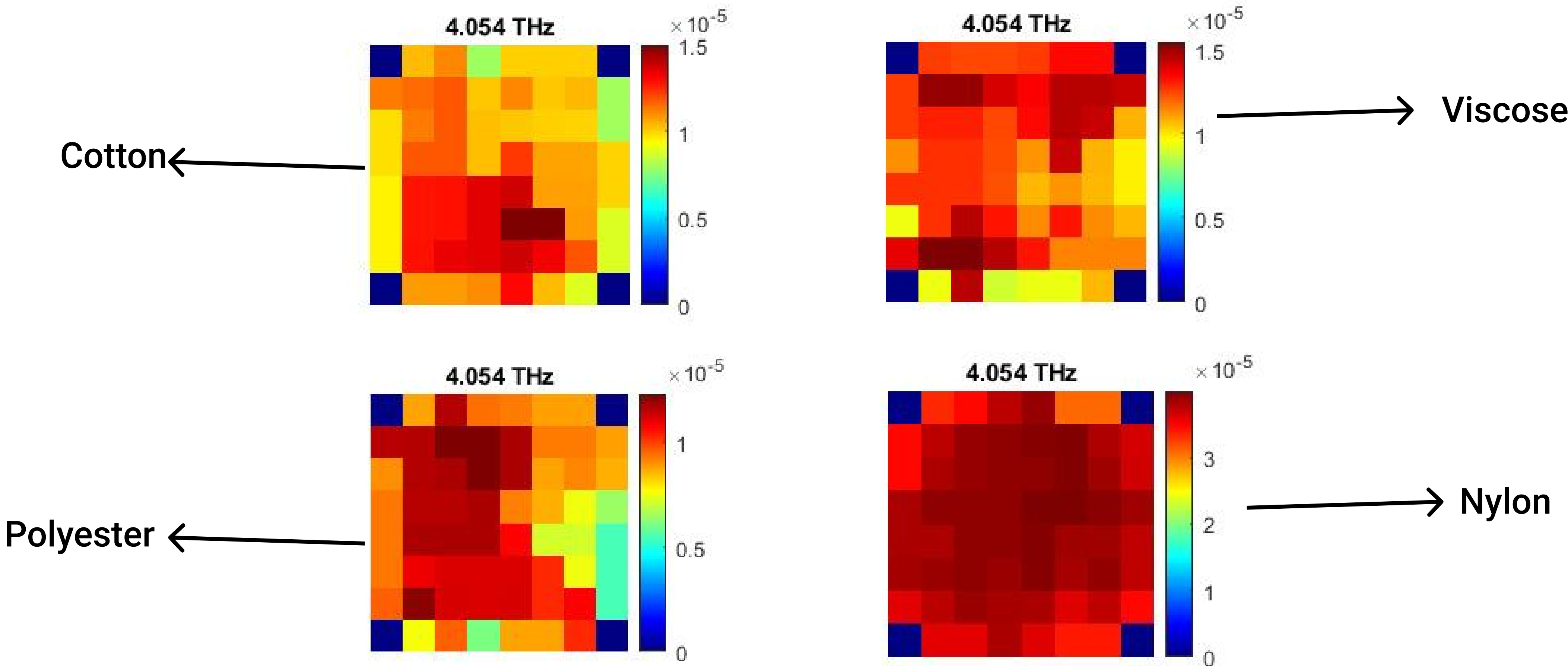
Phase II

Comparative Analysis



Phase II

Comparative Analysis



Hyperspectral Images

Phase II

Comparative Analysis

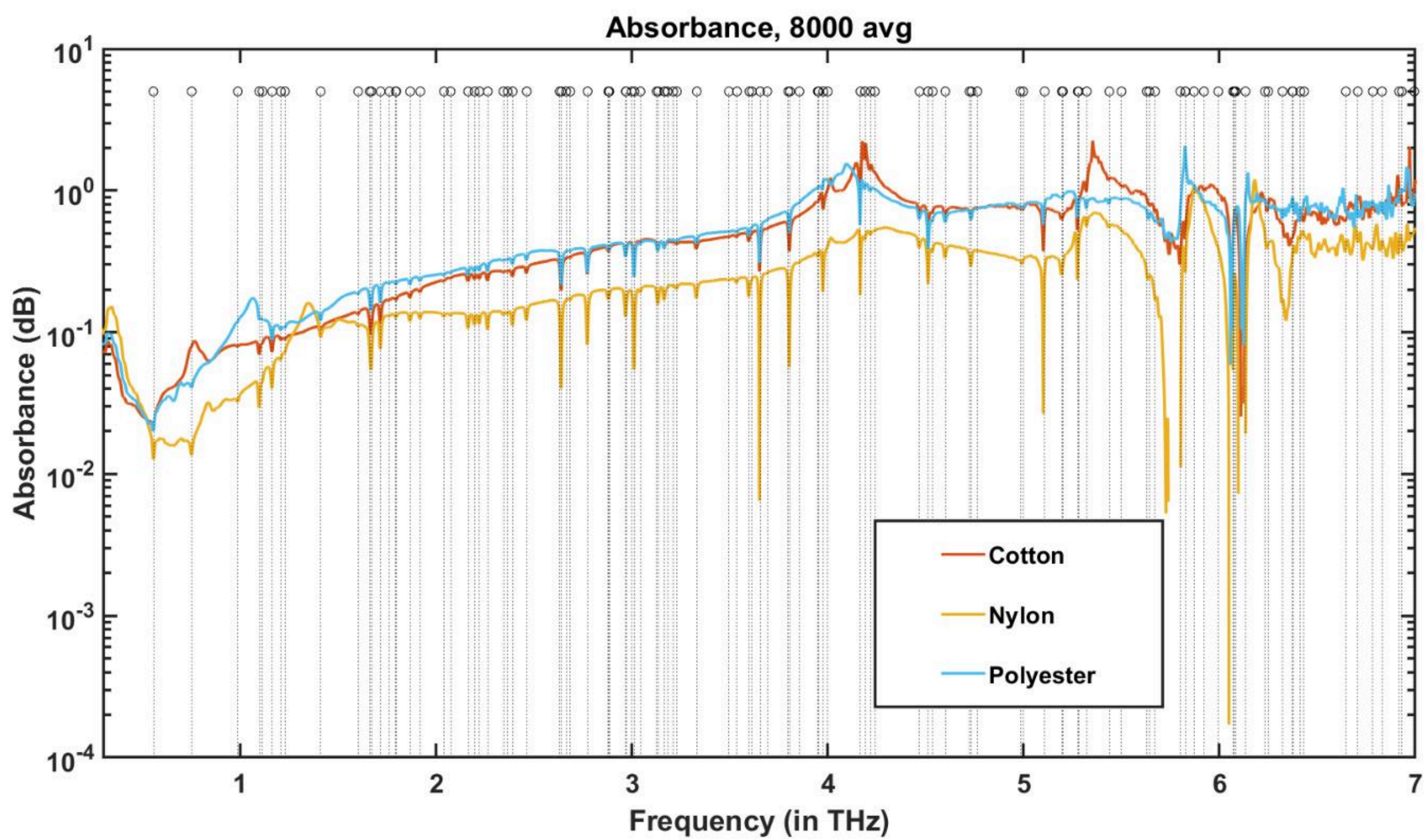


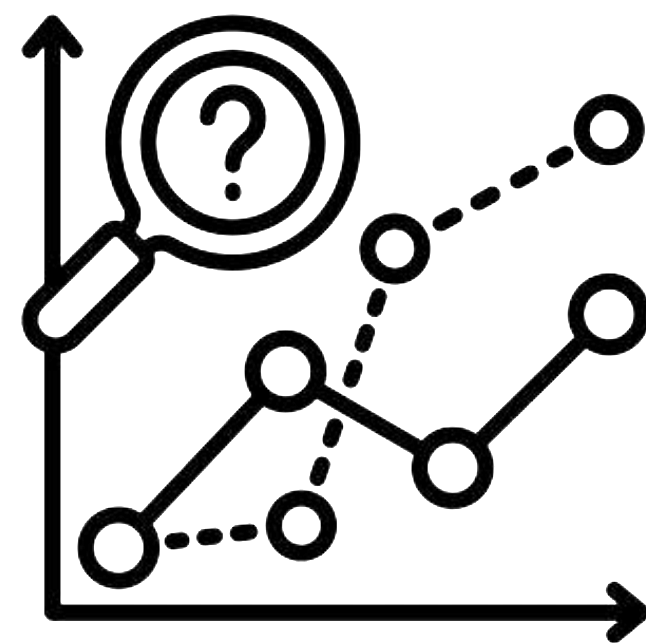
Table 4: Identified Terahertz Absorption Peaks (in THz) for Fiber Samples.

Polyester	Nylon	Viscose	Cotton
1.068	1.346	4.500	0.641
1.254	1.543	0.737	4.238
1.690	4.309	—	0.480

- Samples of nylon, polyester were also used for characterization.
- Distinct spectral signatures identified for each material.
- THz successfully distinguishes chemically similar fibers via structural properties

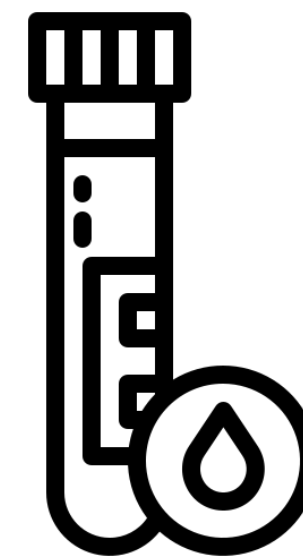
Summary

Gap in Current Research



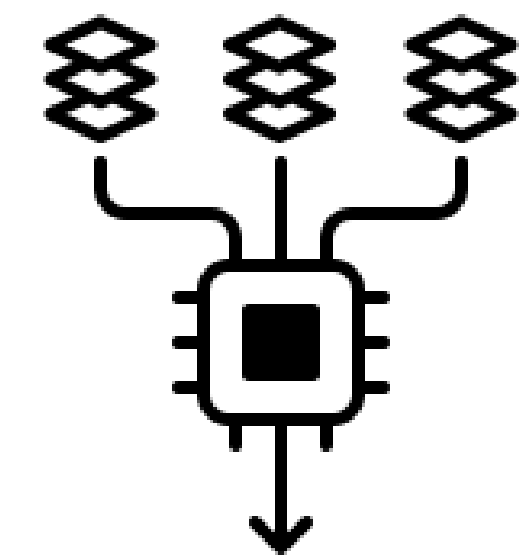
- NIR-based models struggle to distinguish chemically similar fibers.
- Weak NIR Overtones Drive Limited Material Specificity
- Limited exploration of MIR and THz spectroscopy for textiles Fabric.

Progress So Far



- Verified Cotton, Polyester, Nylon using burn tests & chemical solubility tests.
- Created 120 MIR images (40 per fabric type) under controlled conditions.
- Trained YOLO11x-cls architecture on MIR dataset and analysed the results
- Used THZ Spectroscopy Charecterization to improve robustness in response of dye effects.

Next Steps



Sample Expansion

- Grow dataset with more fabric types, blended textiles, and varied conditions.

Model Training & Testing

- Train and benchmark other ML models (KNN, SVM, 1D-CNN).
- Utilize the results of THz spectroscopy for improving identification capabilities of the model.

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